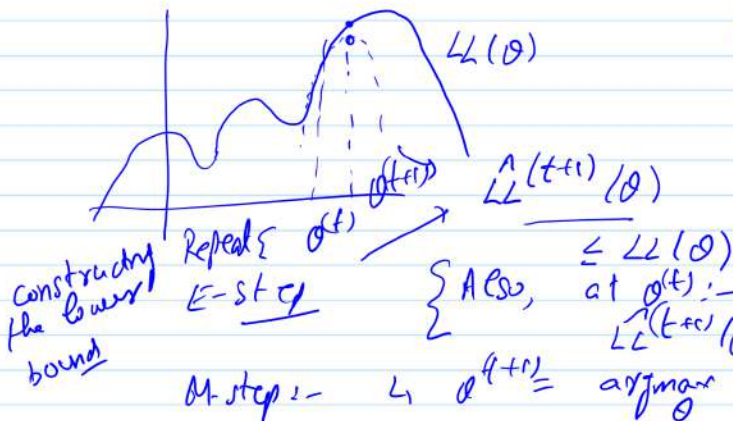


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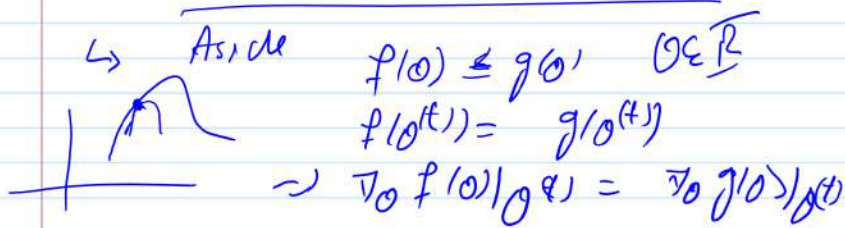


$$L(\theta) = \log \left[\prod_{i=1}^m p(x_i, z_i; \theta) \right] \\ \Rightarrow \sum_{i=1}^m \log \left[\sum_{z_i} p(x_i, z_i; \theta) \right]$$

3 (likely converged) (M-step)

$$L(\theta^{(t)}) \leq L(\theta^{(t+1)}) \leq L(\theta^{(t+2)})$$

By construction



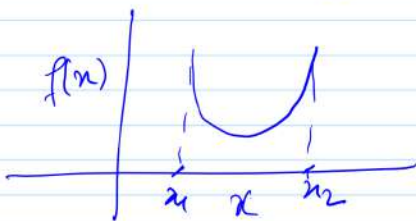
$$\theta^* = \arg \max_{\theta} L(\theta)$$

Suppose sequence converges at $\theta^{(t^*)}$

$$\theta^{(t^*)} = \arg \max_{\theta} L(\theta)$$

$$L(\theta^{(t^*)}) = L(\theta^{(t^*)})$$

Jensen's inequality:-



$p(x)$:- Dist over x
 f : convex function

$$E_{p(x)}[f(x)] \geq f(E_{p(x)}[x])$$

Concave:-

$$f(\alpha x_1 + (1-\alpha)x_2) \geq \alpha f(x_1) + (1-\alpha)f(x_2)$$

if f is strictly concave then:

Concave:-

$$E_{p(x)}[f(x)] \leq f(E_{p(x)}[x])$$

if f is strictly concave then:

$$E_{p(x)}[f(x)] = f(E_{p(x)}[x])$$

$$L(\theta) = \sum_{i=1}^m \log \left[\sum_{z_i} p(x_i, z_i; \theta) \right] = \sum_{i=1}^m \log \left[\sum_{z_i} \frac{p(x_i, z_i; \theta)}{Q(z_i)} Q(z_i) \right]$$

some dist over z_i

Inequality becomes an equality $\Rightarrow \sum_{i=1}^m \sum_{z^u} \left[\log \frac{P(x^u, z^u; \theta)}{Q_i(z^u)} \right] Q_i(z^u) \rightarrow L(\theta)$ (4)

If $\frac{P(x^u, z^u; \theta)}{Q_i(z^u)} = C$

$$\Rightarrow \sum_{z^u} P(x^u, z^u; \theta) = \sum_{z^u} C Q_i(z^u) \Rightarrow C = \sum_{z^u} P(x^u, z^u; \theta) / C = P(x^u; \theta)$$

$$Q_i(z^u) = \frac{P(x^u, z^u; \theta)}{P(x^u; \theta)} = P(z^u | x^u; \theta)$$

Therefore $L(\theta) = \sum_{i=1}^m \sum_{z^u} Q_i(z^u) \log \left[\frac{P(x^u, z^u; \theta)}{Q_i(z^u)} \right] - (5)$

where $Q_i(z^u) = P(z^u | x^u; \theta)$ evaluated $\theta = \theta(t)$

$\Rightarrow \hat{L}(\theta) = L(\theta)$ at $\theta(t)$

$$\Rightarrow \hat{L}(\theta) = \sum_{i=1}^m \sum_{z^u} Q_i(z^u) \log P(x^u, z^u; \theta) + \sum_{i=1}^m \sum_{z^u} (-1) Q_i(z^u) \log Q_i(z^u)$$

$\Rightarrow \arg \max_{\theta} \hat{L}(\theta) = \arg \max_{\theta} \sum_{i=1}^m \sum_{z^u} Q_i(z^u) \left[\log P(x^u, z^u; \theta) \right] - M \text{ step}$

Independent of θ

EM Algorithm

$t \leftarrow 0$

$\theta^{(t)} \leftarrow (\text{init})$

$\sum$

$Q_i^{(t+1)}(z^u) \leftarrow P(z^u | x^u; \theta^{(t)})$ $\rightarrow$ E-step (Assigning missing values)

$\theta^{(t+1)} \leftarrow \arg \max_{\theta} \sum_i \sum_{z^u} Q_i^{(t+1)}(z^u) \log P(x^u, z^u; \theta)$ $\rightarrow$ Maximization step (M-step)

$t \leftarrow (t+1)$

3 (While converged)